

AhamX Bodhi

Where I Becomes Infinite

Level 5: Innovation

Module 2: Generative AI – Core Concepts & Architecture
Prof. Sashikumar Ganesan · IISc Bengaluru



Session Overview

Session Details

- **Module:** 2 – Core Concepts & Architecture
- **Duration:** 10 minutes
- **Bloom's Level:** B3 (Apply)
- **Difficulty:** L2 (Intermediate)
- **Sequence:** Concept 5 of Module 2

The Central Question

Levels 1–4 build, deploy, and scale systems from existing foundation models.

What happens at the frontier – where no foundation model yet exists for your problem, and you must create one?

Level 5 is where the ecosystem itself is advanced. This session maps that territory.

Learning Objectives



By the End of This Session You Will Be Able To

1. **Define** Level 5 Innovation and explain what distinguishes it from Levels 1–4
2. **Identify** the four frontier categories: pre-training, novel architectures, scientific ML, and autonomous systems
3. **Describe** what pre-training a foundation model from scratch requires in terms of data, compute, and infrastructure
4. **Explain** Physics-Informed Neural Networks (PINNs) as a Level 5 scientific innovation pattern
5. **Recognise** real-world Level 5 landmark systems – Bloomberg GPT, AlphaFold 3, Llama 3.1 405B, FutureHouse



Prerequisites and Context

You Should Already Know

- Foundation models and self-supervised pre-training (Level 1)
- The three adaptation strategies: prompting, RAG, fine-tuning (Level 2)
- AI agent architecture and the ReAct loop (Level 3)
- LLMOps, serving, and production integration (Level 4)
- Scaling laws – Kaplan et al. (2020) and Chinchilla (2022)

This Session Unlocks

- Scientific ML and PINNs – Module 6
- Pre-Training Custom Models – Module 8
- Advanced Multi-Agent Systems – Module 9
- AI Safety and Alignment Research – Module 9
- Capstone: Novel Production System – Module 10





What Is Level 5 Innovation?

Definition

Level 5 Innovation is the creation of fundamentally new AI capabilities that did not previously exist – through pre-training new foundation models from scratch, designing novel neural architectures, encoding domain knowledge directly into training objectives, or building autonomous systems that advance the state of the art. Level 5 work produces artefacts that **the entire ecosystem at Levels 1-4 then builds upon**. The outputs of Level 5 are tomorrow's foundation models.



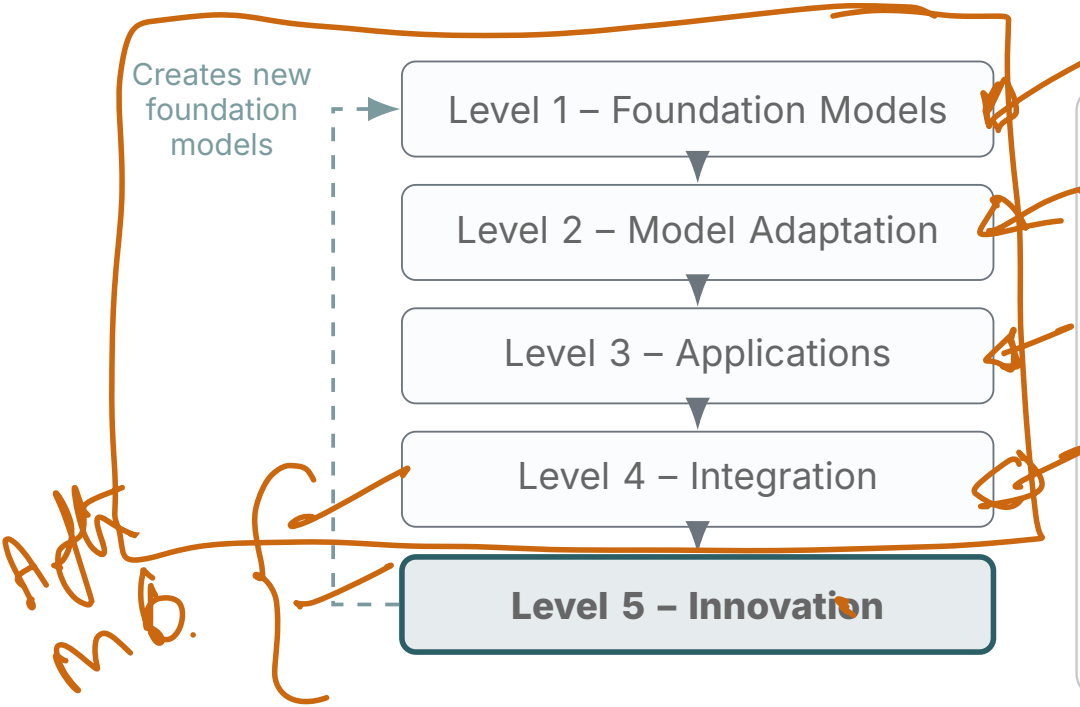
What Is Level 5 Innovation?

The Four Level 5 Frontiers

- **Foundation Model Pre-Training:** Training billion-to-trillion parameter models on web-scale data from scratch
- **Novel Architecture Design:** Inventing new attention mechanisms, loss functions, and model families
- **Scientific ML:** Physics-Informed Neural Networks, Neural Operators, neurosymbolic systems
- **Autonomous Innovation Systems:** Self-improving agents, AI co-scientists, recursive research pipelines



Level 5 in the Full Framework Stack



The Unique Property of Level 5

- Levels 1–4 are **consumers** of foundation models: they adapt, apply, and deploy them
- Level 5 is the **producer**: it creates the foundation models and architectural innovations
- This creates a feedback loop – Level 5 outputs become Level 1 inputs for the entire ecosystem
- Every GPT, Claude, Llama, and Gemini release is a Level 5 event





Pre-Training a Foundation Model: What It Takes

The Scale of Pre-Training

Pre-training a frontier foundation model is a **multi-month, multi-hundred-million-dollar engineering programme**. Meta's Llama 3.1 405B was trained on 15.6 trillion tokens across 16,000 NVIDIA H100 GPUs for several months. Google's Gemini Ultra required custom TPU v5e pods. OpenAI's GPT-4 reportedly cost over \$100 million to train. This is not a single researcher's laptop project – it is a coordinated systems-engineering effort.



Pre-Training a Foundation Model: What It Takes

The Five Pre-Training Requirements

- **Data:** Trillions of high-quality tokens – web crawls (Common Crawl), books, code, papers, curated domains
- **Tokenizer:** Custom BPE/SentencePiece tokenizer trained on the full corpus before model training
- **Architecture:** Decoder-only transformer with chosen hyperparameters (layers, heads, d_{model} , context length)
- **Compute:** Thousands of A100/H100 GPUs coordinated with pipeline and tensor parallelism
- **Training infrastructure:** Megatron-LM, DeepSpeed, PyTorch FSDP, SLURM cluster, checkpoint management



Scaling Laws: The Blueprint for Pre-Training Decisions

Chinchilla Optimal Allocation

- Hoffmann et al. (2022, DeepMind): given a fixed compute budget C , optimal training balances model size N and data D
- Rule of thumb: ≈ 20 tokens per parameter
- GPT-3 (175B params) was **under-trained** – only 300B tokens
- Chinchilla (70B) on 1.4T tokens **outperforms** GPT-3 on most benchmarks at $10\times$ lower parameter count

Llama 3 Applies Chinchilla

- Llama 3 8B trained on 15T tokens (1,875 tokens/parameter – far beyond Chinchilla optimum for inference efficiency)
- Over-training a smaller model is **strategically optimal**: cheaper inference at deployment
- This insight drives the trend toward smaller, heavily-trained models (Phi-3, Gemma 2, Llama 3.2)
- Level 5 practitioners must reason about **training compute vs. inference cost** trade-offs





Key Architectural Innovations at Level 5

Year	Innovation	Authors	Impact
2017	Transformer (Attention Is All You Need)	Vaswani et al. (Google)	Foundation of all modern LLMs
2020	Sparse Mixture-of-Experts (MoE)	Lepikhin et al. (Google)	Decouples parameters from compute cost
2022	FlashAttention	Dao et al. (Stanford)	2–4× faster training; enables long context
2023	Grouped Query Attention (GQA)	Ainslie et al. (Google)	Faster inference, lower VRAM for large models
2024	Multi-Head Latent Attention (MLA)	DeepSeek-AI	5–13× KV cache reduction at inference
2024	Test-Time Compute Scaling	OpenAI o1 team	Inference-time reasoning via chain-of-thought



Designing Novel Architectures: The Research Process

The Architecture Research Loop

- Identify a **fundamental bottleneck** (memory, speed, reasoning depth, context length)
- Formulate a mathematical hypothesis (e.g. low-rank KV compression)
- Implement and ablate at small scale (125M–1B parameters)
- Scale up to verify the hypothesis holds (7B–70B)
- Publish, open-source, and integrate into the ecosystem

What This Requires

- Deep mastery of transformer internals: attention, normalisation, positional encoding, MLP blocks
- Proficiency in PyTorch custom CUDA kernels and Triton for GPU-level optimisation
- Access to ablation compute – dozens of small training runs
- Research community engagement – paper submission, peer review, replication
- Tools: Megatron-LM, DeepSpeed ZeRO, Weights & Biases, SLURM job schedulers





Physics-Informed Neural Networks (PINNs)

What Are PINNs?

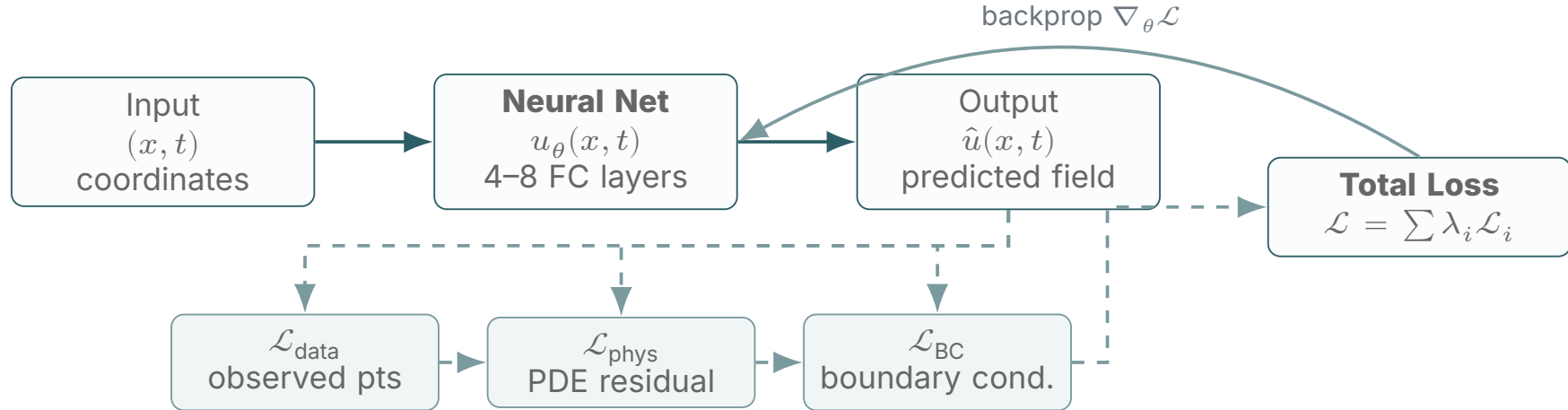
A **Physics-Informed Neural Network** is a neural network trained to satisfy both observed data **and** the governing physical equations of a system simultaneously. Instead of learning purely from data, the model's loss function encodes the **partial differential equations (PDEs)** that describe the physics – conservation of mass, momentum, energy, or heat. The network is forced to produce solutions that are physically plausible even in regions with no training data.

The PINN Loss Function

- **Data loss** $\mathcal{L}_{\text{data}}$: standard MSE between predictions and observed measurements
- **Physics loss** $\mathcal{L}_{\text{phys}}$: residual of the PDE evaluated at collocation points – e.g.
 $\|\partial_t u + u \cdot \nabla u - \nu \nabla^2 u\|^2$ for Navier-Stokes
- **Boundary loss** $\mathcal{L}_{\text{BC}}$: enforces boundary and initial conditions
- **Total loss**: $\mathcal{L} = \lambda_1 \mathcal{L}_{\text{data}} + \lambda_2 \mathcal{L}_{\text{phys}} + \lambda_3 \mathcal{L}_{\text{BC}}$



PINN Architecture: Encoding Physics into the Loss



PINNs require **automatic differentiation** (PyTorch/JAX) to compute PDE residuals through the network – a capability unique to deep learning frameworks.



Beyond PINNs: The Scientific ML Frontier

Neural Operators

- PINNs solve one PDE instance per training run; **Neural Operators** learn the solution *operator* – mapping any PDE input to its solution
- **FNO (Fourier Neural Operator)**: operates in Fourier space; $1,000\times$ faster than classical solvers for fluid dynamics
- **DeepONet**: learns branch-trunk architecture to approximate any operator
- Tools: DeepXDE, NVIDIA Modulus, JAX-MD

Scientific ML Domains

- **Climate modelling**: Neural surrogates replace expensive GCM components; GraphCast (DeepMind) outperforms ECMWF at 10-day forecasting
- **Drug discovery**: AlphaFold 3 predicts protein–ligand structures; ESM-2 generates novel protein sequences
- **Materials science**: GNoME (DeepMind, 2023) discovered 2.2 million new stable crystal structures
- **Plasma physics**: DeepMind's MagnetoHydrodynamics control at JET tokamak





Autonomous Innovation: AI Co-Scientists

The Emerging Paradigm

AI co-scientist systems are autonomous agents designed to conduct the full scientific research loop: literature review, hypothesis generation, experiment design, execution (via tool use), result analysis, and paper writing – with minimal human direction per cycle. Unlike ReAct agents (Level 3) that assist a user with a defined task, co-scientists **set their own research sub-goals** across long time horizons. This is nascent but advancing rapidly.



Autonomous Innovation: AI Co-Scientists

Production Co-Scientist Systems (2024–2026)

- **FutureHouse AI:** Platform of specialised science agents – Crow (literature), Falcon (deep search), Owl (deduplication), Phoenix (chemistry) – working in coordination
- **Google DeepMind AI co-scientist:** Gemini-based system that proposes novel research hypotheses in biomedicine; validated against expert-written proposals
- **Sakana AI:** “AI Scientist” system that generates, experiments, reviews, and writes ML papers autonomously
- **OpenAI o3 Deep Research:** Multi-hour reasoning agent that produces PhD-level literature synthesis reports





Application: Bloomberg GPT -- Domain Pre-Training

What It Is

- 50-billion-parameter LLM pre-trained from scratch by Bloomberg (2023)
- Trained on a 700-billion-token mixture: 363B tokens of financial text (FinPile) + 345B tokens of general web/book text
- Purpose: capture financial domain knowledge – earnings reports, analyst notes, regulatory filings, market terminology – that general LLMs handle poorly
- Published as a research paper; model not publicly released

Why It Is Level 5

- Adapting GPT-4 via prompting or fine-tuning (Levels 2–3) could not match domain-specific pre-training for NER, sentiment, and numeric reasoning in finance
- Required custom data pipeline: filtering, deduplication, and tokenizer extension for financial symbols
- Outperforms GPT-4 on financial NLP benchmarks (FPB, FiQA, NER) while being 3× smaller
- Demonstrates the case for **industry-specific foundation models**



Application: AlphaFold 3 -- Scientific Architecture Innovation

The Breakthrough

AlphaFold 3 (Google DeepMind, 2024) is a diffusion-based transformer that jointly predicts the 3D atomic structure of proteins, DNA, RNA, small molecules, and their complexes – from sequence alone. It replaced AlphaFold 2's MSA-based pipeline with a **pairwise diffusion architecture** (*Evoformer* → *Pairformer* + *Diffusion Module*), achieving state-of-the-art accuracy on protein-ligand docking and nucleic acid structure prediction simultaneously. This is a pure Level 5 architectural innovation.



Application: AlphaFold 3 -- Scientific Architecture Innovation

Scientific and Industry Impact

- Predicts drug-target binding poses – reducing early-stage drug discovery from years to hours
- Enables rational vaccine design: modelling antigen-antibody complex geometry at atomic resolution
- Insilico Medicine used AlphaFold predictions to design two clinical drug candidates in 18 months
- AlphaFold DB now covers >200 million predicted protein structures, freely accessible to researchers



Application: Llama 3.1 405B -- Open-Source Frontier Pre-Training

The System

- Meta FAIR pre-trained a 405-billion-parameter dense transformer on 15.6 trillion tokens
- 128K token context window; multilingual; released with open weights
- Training infrastructure: 16,000 NVIDIA H100 GPUs, pipeline + tensor + data parallelism via Megatron-LM
- Post-training: supervised fine-tuning, reward modelling, DPO alignment for Llama 3.1 Instruct

Why It Matters for the Ecosystem

- First open-weight model competitive with GPT-4o on major benchmarks
- Enables **sovereign AI**: organisations can fine-tune, deploy, and serve a frontier model on their own infrastructure
- Accelerated the open-source community: Llama 3.1 fine-tunes (Hermes, Dolphin, WizardLM) emerged within days of release
- Demonstrates that Level 5 pre-training need not be exclusive to closed labs



Application: Your Path to Level 5 -- The IISc Programme

How This Programme Bridges to Innovation

Most practitioners will never pre-train a GPT-4-scale model. But **Level 5 contribution is not binary**. Publishing a novel PINN loss formulation for a new PDE is Level 5. Pre-training a 1-billion-parameter domain model for your industry is Level 5. Building a multi-agent system that autonomously discovers drug candidates is Level 5. The IISc programme is designed to bring you to the threshold of contribution – not just consumption.

Module 8 and 10 – Your Level 5 On-Ramp

- **Module 8:** Pre-train a custom $<1B$ parameter language model from scratch – data pipeline, tokenizer, causal LM training, scaling law experiments
- **Module 6:** Implement PINNs for a real PDE – heat transfer, fluid flow, or wave equations – using DeepXDE
- **Module 10 Capstone:** 30 themes available; several are explicitly Level 5 – novel agent architectures, domain pre-training, scientific ML systems



Knowledge Graph: Where This Concept Lives

Prerequisite Concepts

- Level 1: Foundation Models (Module 2)
- Level 2: Model Adaptation (Module 2)
- Level 3: Applications and Agents (Module 2)
- Level 4: Integration and LLMOps (Module 2)
- Model Sizing and Scaling Laws (Module 1)

Concepts This Unlocks

- Scientific ML and PINNs – Module 6
- Pre-Training Custom Models – Module 8
- Advanced Multi-Agent Architecture – Module 9
- AI Safety and Alignment Research – Module 9
- Capstone: Novel System Design – Module 10

Common Misconceptions



Critical Corrections

- **"Level 5 is only for researchers at big labs"** – Any practitioner who pre-trains a domain model, publishes a novel architecture ablation, or builds a co-scientist agent is operating at Level 5. Scale is not the only axis of innovation
- **"Fine-tuning on enough data equals pre-training"** – Fine-tuning updates weights relative to an existing distribution; pre-training from scratch shapes the entire representational geometry. They are architecturally and epistemically distinct
- **"PINNs are just regular neural networks with extra loss terms"** – The physics loss requires computing PDE residuals via automatic differentiation through the network, which changes the training landscape, convergence dynamics, and data requirements fundamentally
- **"Novel architectures must be dramatically different from Transformers"** – Most impactful innovations (FlashAttention, GQA, MLA) are targeted modifications to specific bottlenecks, not wholesale replacements. Incremental architectural research is legitimate Level 5 contribution
- **"AI co-scientists will replace human researchers"** – Current systems excel at literature retrieval and hypothesis breadth but lack the creative leaps and contextual judgment of experienced domain scientists. They are co-pilots, not replacements



Module 2 Complete: The Full Five-Level Architecture

Level	Stage	Core Capability	Key Tools	Programme
L1	Foundation Models	Self-supervised pre-training; emergent capabilities	GPT-4, Llama, Claude	M1, M3, M8
L2	Model Adaptation	Prompting, RAG, fine-tuning; task specialisation	LangChain, PEFT, LoRA	M2, M4, M7
L3	Applications	Agent loops, tool use, ReAct, MCP, multi-agent	LangGraph, vLLM, FastAPI	M5, M9
L4	Integration	LLMOps, serving, monitoring, cost control, safety	Docker, Prometheus	M9, M10
L5	Innovation	Pre-training, novel architectures, PINNs, co-scientists	Megatron, DeepXDE	M6, M8, M10

Session Summary



Key Takeaways

- Level 5 creates foundation models and architectural innovations that power the entire ecosystem at Levels 1–4
- The four frontiers are: pre-training, novel architectures, scientific ML (PINNs/Neural Operators), and autonomous co-scientist systems
- Chinchilla scaling laws govern the optimal allocation of compute between model size and training data
- Level 5 contribution does not require billion-dollar budgets – a novel PINN formulation or a domain-pre-trained model qualifies

What Comes Next in the Program

- **Module 3:** Transformer architecture deep-dive – the foundation every level builds on
- **Module 6:** Implement PINNs for real PDEs using DeepXDE – your first Level 5 hands-on work
- **Module 8:** Pre-train a <1B parameter LM from scratch – data, tokenizer, training loop
- **Module 10:** Capstone – choose a Level 5 theme and build a production innovation system

Thank You

Module 2: Generative AI – Core Concepts & Architecture
Level 5: Innovation · Prof. Sashikumaar Ganesan · IISc Bengaluru